

BEC305 Python Programming — Assignment 3 —

Regular expressions and file handling

Module 3: Pattern Matching with Regular Expressions and File Handling

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	<<TBD>>
Due date	<<TBD>>
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO3: 13 · L1–L2 0, L3–L4 8, L5–L6 5

Objective

Develop the ability to describe a shape of text precisely and to move data between memory and disk safely, with particular attention to greediness, anchoring, path portability and the destructive behaviour of write mode.

Learning outcomes

- Student will be able to write, compile and apply regular expressions using groups, quantifiers and character classes.
- Student will be able to distinguish searching from validating and anchor patterns accordingly.
- Student will be able to build portable paths and inspect the filesystem before opening a file.
- Student will be able to persist a Python structure and recover it, choosing between shelve and pformat with a stated reason.

Part A — Problems (17 marks)

P1. Write the phone-number matcher by hand with explicit checks, then replace it with a regular expression. Count the lines in each, then change the accepted format and report how many lines changed in each version. [C03 | L4 | 2 marks]

P2. Write a pattern with at least three capturing groups that extracts date parts from a text. Demonstrate group(0), each individual group, and groups() unpacked by multiple assignment. [C03 | L3 | 2 marks]

P3. Construct an alternation in which one alternative is a prefix of another. Show the truncated match, reorder to fix it, and explain why no error was raised. [C03 | L4 | 2 marks]

P4. Demonstrate ?, *, + and {n,m} on the same base pattern, testing each against input with zero, one and several occurrences. Tabulate the nine results. [C03 | L4 | 2 marks]

P5. Show the greedy and nongreedy forms of one pattern producing different matches on multi-occurrence input, and identical matches on single-occurrence input. State the testing rule this implies. [C03 | L4 | 2 marks]

- P6.** Write a validator for a username or a postcode of your choice. Show that the unanchored version wrongly accepts contaminated input and that the anchored version rejects it. [C03 | L4 | 2 marks]
- P7.** Write a phone-and-email extractor using re.VERBOSE, with the pattern laid out over several commented lines. Run it against a block of text and report every match found. [C03 | L5 | 2 marks]
- P8.** Write a program that walks a folder, totals the size of its files, and reports the largest. Use os.path.join throughout and guard every access with an isfile check. [C03 | L3 | 1 marks]
- P9.** Demonstrate all three file modes on the same file, recording its contents after each step. Show explicitly how 'w' inside a loop destroys all but the last write. [C03 | L4 | 1 marks]
- P10.** Persist the same nested structure twice, once with shelve and once with pprint.pformat into an importable .py file. Recover both and state which you would choose for this data and why. [C03 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Explain why a regular expression moves a specification from control flow into data, and what that buys over the life of a program. [C03 | L5 | 1 marks]
- T2.** Argue why findall changing its return type when a group is added is a reasonable design, and describe the discipline it requires of the caller. [C03 | L5 | 1 marks]
- T3.** Explain why 'w' mode truncates on opening rather than on the first write, and what a program must do to protect data that matters. [C03 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC305_Assignment3.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Pattern construction and correctness	7	Every pattern a raw string, groups and quantifiers used precisely, all results verified against the stated inputs.	Patterns correct with one minor quantifier or grouping slip.	Patterns work on the given input but are fragile on variations.	Patterns not raw strings, or group numbering misunderstood.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Anchoring, greediness and return shapes	5	Search versus validate distinguished explicitly, greedy and lazy contrasted on multi-occurrence input, findall return shape stated for every call.	Correct treatment with one gap.	Concepts demonstrated but not explained.	Validator left unanchored, or tuple results treated as strings.
File and path handling	5	os.path.join used throughout, every access guarded, with statement used for all file work, modes chosen deliberately.	Sound handling with occasional unguarded access.	Works on the author machine only.	Hardcoded separators, or data destroyed by an unintended w mode.
Persistence choice and justification	3	Both mechanisms demonstrated, round trip verified, and the choice justified against who must read the data.	Both demonstrated with a brief justification.	One mechanism demonstrated.	Persistence attempted without verifying the round trip.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO3	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.